

Data Cleaning & Preparation — Change Log

Project 1 | DecodeLabs Industrial Training Kit | Batch 2026

Dataset Summary

Source file: Dataset_for_Data_Analytics.xlsx — 1,200 order records, 14 columns (OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice).

Audit Checks Performed

Check	Result
Duplicate OrderIDs	0 found
Fully duplicate rows	0 found
Missing values (all columns)	309 missing in CouponCode only
Date format consistency (ISO 8601 parseable)	Pass — all 1,200 rows valid
TotalPrice = Quantity x UnitPrice	Pass — 0 mismatches
Negative/zero Quantity, UnitPrice, ItemsInCart	0 found
ID format consistency (OrderID, CustomerID, TrackingNumber)	Pass — 100% match expected pattern
Categorical text consistency (Product, PaymentMethod, OrderStatus, ReferralSource)	Pass — no spelling variants found

Changes Made

Change ID	Description	Impact	Status
CR001	Imputed 309 missing CouponCode values with "NONE"	Revised all 1,200 records; no rows Deleted	Completed

Rationale

CouponCode gaps were treated as "no coupon applied at checkout" rather than an unknown/missing measurement, since every other order shows a valid code (SAVE10, FREESHIP, or WINTER15) and there is no indication these 309 orders failed to record data. Listwise deletion was avoided per data-integrity standards, since removing these rows would reduce statistical power and discard otherwise valid order records. An explicit label ("NONE") was used instead of leaving blanks, so downstream analysis can distinguish "no coupon used" from "data missing."

Verification Gate (Threshold for Project 2)

Duplicate ID error rate: 0% — PASS
Incorrect date format error rate: 0% — PASS